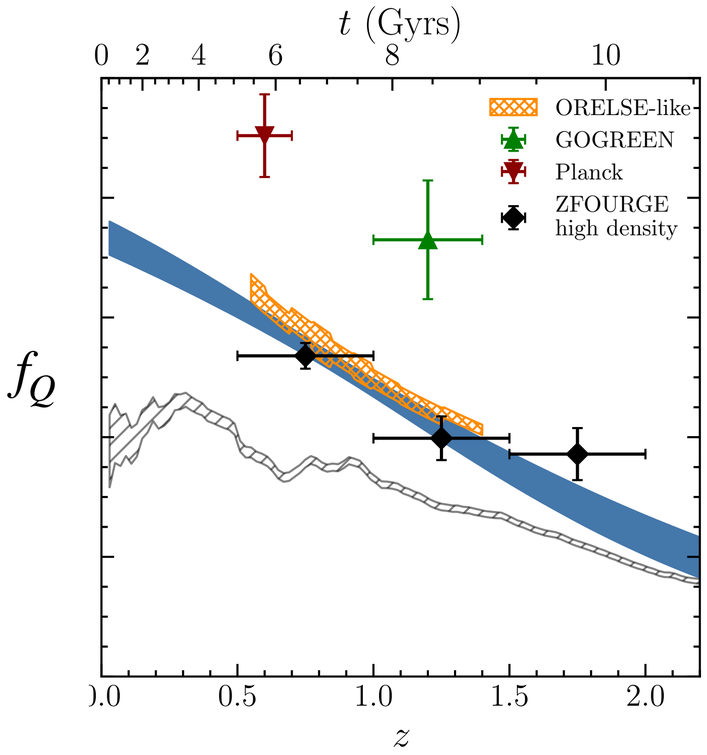
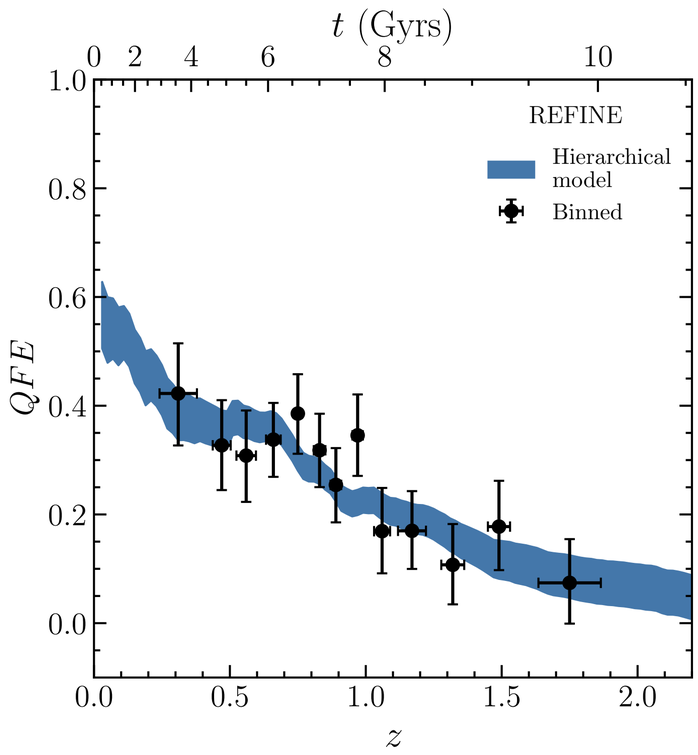
Results from REFINE

Sarron & Conselice (2021)

Stellar mass limits : $10.25 < \log M_\star/\mathrm{M}_\odot < 11$



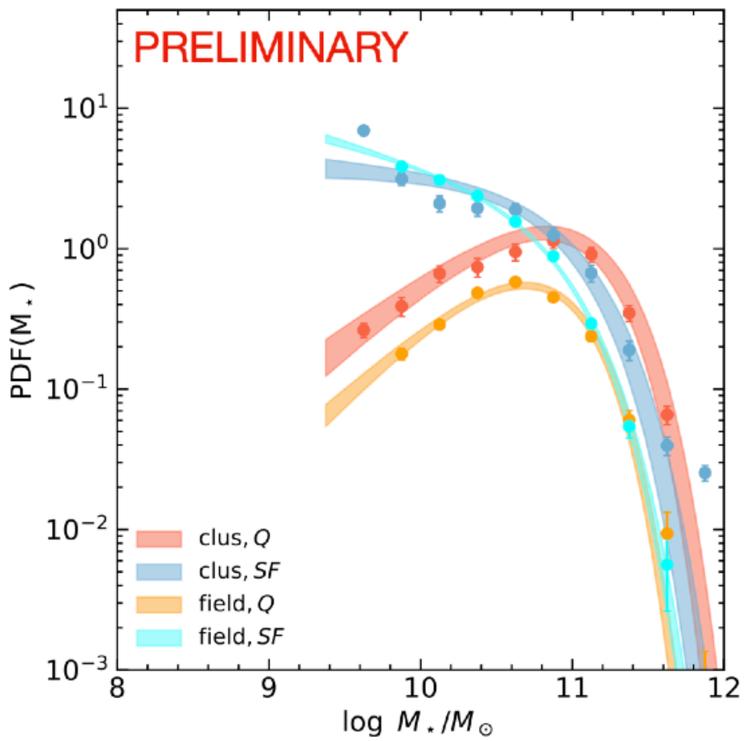
$$QFE = \frac{f^q_{\mathrm{group}} - f^q_{\mathrm{field}}}{1 - f^q_{\mathrm{field}}}$$



Galaxy stellar-mass functions

New method : use SED fitting $\mathrm{PDF}(z, M_\star)$ and $p_Q(z, M_\star)$ in the inference of cluster galaxy stellar-mass functions

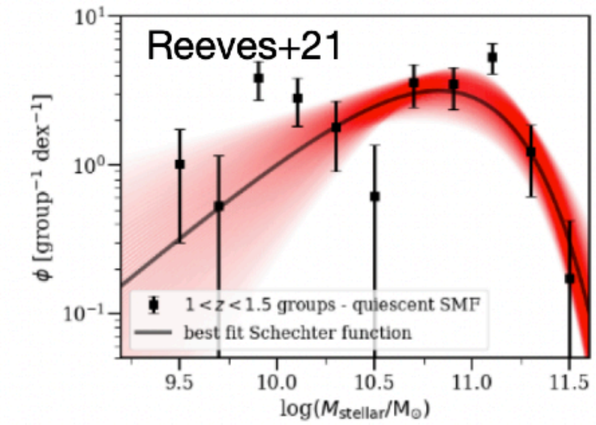
→ important for Euclid/LSST samples were uncertainty will be dominated by systematics



Application to REFINE

- Competitive constraints for the group environment at $z > 1$

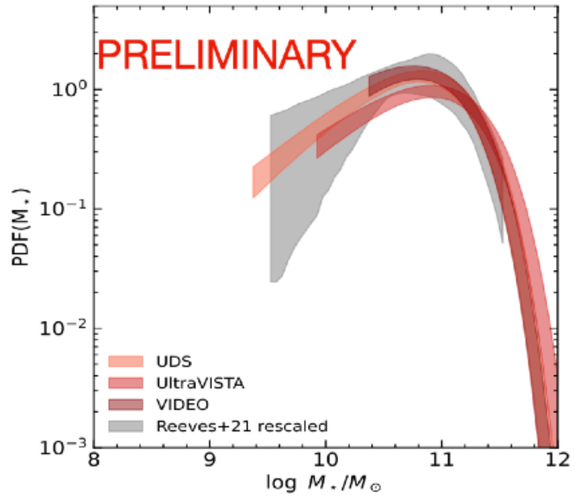
$1 < z < 1.5$



GOGREEN

21 groups

X-ray selected



REFINE

328 candidate groups

$\mathrm{purity}(z) > 50\,\%$